

Those who apply it widely will derive the maximum value from it. Today, AI and algorithms are transforming the very nature of our society. They are being used across sectors such as agriculture, healthcare, education, manufacturing, transport, governance and climate action.

In agriculture, for instance, AI-based sowing advice has helped farmers in Andhra Pradesh increase crop yields by up to 30 per cent. AI-powered decision-making in Maharashtra has helped farmers increase sugarcane yield by 50 per cent. AI is also helping improve milk yield and animal health. Wearable IoT (internet of things) devices now track nutrition, fertility and early signs of disease in farm animals. In several states, these solutions have reduced methane emissions by up to 30 per cent.

In healthcare, AI solutions are bringing top-notch hospital capabilities to district hospitals. Indian startups are developing AI screening tools that support doctors by analysing medical images and other clinical data. AI is also helping detect neurological diseases earlier and more accurately by deciphering complex brain signals. It is clear that any nation that wants to grow rapidly must learn to use AI at scale across sectors. We are following this strategy of 'AI diffusion' or AI adoption at scale.

India is among the top three startup ecosystems. Today, it is estimated that nearly 90 per cent of startups are AI-powered in some form. This shows how deeply AI is already embedded in innovation.

> INDIA'S DIGITAL REVOLUTION PHILOSOPHY IS **THAT TECHNOLOGY MUST BE DEMOCRATISED**, BOTH IN USAGE AND DEVELOPMENT. **AI, TOO, WILL FOLLOW THE SAME PATH**

AI Models: Brain Behind the Applications

The second layer of the stack is AI models. While applications create visible societal impact, the AI model serves as the brain that powers them. Large global players, including OpenAI, Google and Meta, led the early progress in developing these frontier models. Those models demonstrated what AI could achieve at scale and expanded the boundaries of what was technologically possible.

However, training and running these models required enormous computing resources, making them expensive to develop and deploy. As a result, access to advanced AI development capabilities remained limited to a few players. Then came engineering innovations that brought a slew of models from China. Decades of investment in engineering and technological research propelled these innovations.

As the model layer evolves, we have a very large number of open-source models that are developed in so many countries. These models lower the entry barriers by reducing the cost of training and deployment.

Through these models, startups, researchers and developers can build on existing work rather than starting from scratch. This makes innovation faster, improves transparency and builds trust. It also allows AI to be tailored to local languages, specific sectors, and also adapt to national regulations. But in the age of AI, it is imperative for every nation to have its sovereign models too, for they ensure data security, cultural relevance and strategic autonomy. To achieve this goal, 12 indigenous AI models are being developed under the IndiaAI Mission. These models will be used to solve real-world problems at scale in the Indian context.

One example is Sarvam AI, which is building full-stack, India-first AI models in multiple Indian languages. It will help users for voice calls, documents and citizen services, understanding and responding in Indian languages. These models have been trained from scratch in 22 Indian and six UN languages.

In healthcare, the Qure.ai model uses AI to assist doctors in detecting TB, lung cancer and other conditions from medical images, reaching millions globally. NeuroDx is using AI to analyse brain signals (EEG) to detect neurological conditions like epilepsy and dementia early. This will enable specialised services in remote hospitals. BharatGen is developing open, India-centric foundation models in Indian languages, with 2 billion to 1 trillion parameters. These models will support research, startups and public sector applications. This shows how Indian startups are using AI to deliver af-